

2024



Reducing Anzu’s Cloud Costs by 50% on AWS

Learn how intrinsic in-game advertising company Anzu unlocked better performance at a lower cost by migrating to AWS.

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30%

increase in daily bid requests

50%

reduction in monthly cloud costs

5
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reduction in carbon footprint

Overview

When intrinsic in-game advertising company [Anzu](#) faced networking and service issues from its cloud services provider, it knew that making a change would empower it to deliver higher-quality service to its users. By migrating to Amazon Web Services (AWS), Anzu unlocked performance improvements, cost savings, and a reduced carbon footprint, which created a better experience for both its internal teams and its customers.



Opportunity | Using AWS Services to Improve Performance for Anzu

Founded in 2017, Anzu is an advertising company that operates in the gaming industry. Its mission is to make advertising in games better—for brands, developers, and gamers. The company achieves this through its intrinsic in-game advertising solution, which helps advertisers reach gamers across mobile, PC, and console games with ads that are integrated into the game world, such as on in-game products and billboards. Using Anzu’s solution, game developers can monetize their product through advertising without damaging player immersion or interrupting the experience with intrusive or out-of-place allurements.

From its inception Anzu has had rapid expansion. It now works with major companies in the gaming industry, including Ubisoft, Gameloft, Saber Interactive, and Socialpoint, and reaches more than 150 million users per month. As a result of its growth, Anzu started having networking, load balancing, and availability issues with its original cloud services provider. After going through three major events within 1 year, Anzu sought a cloud provider that could offer enhanced performance and greater capabilities. Additionally, the company wanted to improve cost-efficiency.

In January 2024 Anzu began a 3-month migration to AWS. The company anticipated that the enhanced stability, improved uptime, and better service-level agreements of AWS services would improve performance both internally and for its end customers. “We really liked the AWS offerings,” said



cofounder and chief product officer at Anzu. “Overall, AWS provided more capabilities and better services. It’s more mature than our previous solution.”

Anzu completed its migration with support from the AWS team—which involved rebuilding and enhancing its entire programmatic advertising infrastructure on AWS. Anzu carefully managed the pace of migration, which facilitated a seamless transition for users without interrupting ongoing operations. “Migration was so straightforward that we went from initial planning to full production in just 90 days,” says Fenster. “It was a really smooth experience.”

Solution | Cutting Monthly Cloud Costs by 50 Percent

Anzu’s solution processes 25 billion near real-time bidding requests and serves ads to tens of millions of users each day. At this scale, dynamic and flexible networking is essential for maintaining high-quality service. The company now uses [Amazon CloudFront](#), which securely delivers ad content with low latency and high transfer speeds. To manage all incoming traffic, Anzu has adopted [Elastic Load Balancing](#), which automatically distributes incoming application traffic across multiple targets in one or more Availability Zones. With these enhanced networking and load-balanced solutions in place, Anzu increased its bid request capacity by 30 percent and reduced ad auction latencies by 20 percent compared with its previous network solution, thereby significantly improving performance. “We can get powerful machines, but in the end, what will limit them is the amount of network throughput they can push,” says Fenster. “It’s important to our business to have the highest-quality networking.”

Anzu powers its solutions with managed services, including [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service to run Kubernetes in the AWS Cloud. The company’s bidders, ad gateway, and ad quality scanners use [Amazon Elastic Compute Cloud](#) (Amazon EC2) [Spot Instances](#), which let developers take advantage of unused Amazon EC2 capacity in the AWS Cloud and are available at up to a 90 percent discount compared with On-Demand prices. Anzu used [Karpenter](#), a flexible, high-performance Kubernetes cluster autoscaler, to better manage its workloads and improve cost-efficiency. The combination of these services helped Anzu achieve a 50 percent reduction in both machine startup time and compute reservation time—the time it takes for a new instance to be ready for use. “Lower machine reservation time matters a lot because if your infrastructure is stressed, taking 5 minutes to react or taking 30 seconds to react can mean the difference between the infrastructure failing or not,” says Fenster. Having fewer machine reserves led to substantial cost savings for Anzu. In fact, the company reduced its operational overhead and cut its monthly cloud costs, particularly in data transfer, by 50 percent, which translated to over 1 million dollars in 2024 alone.

Migrating to AWS also empowered Anzu to cut down the number of servers it needed to handle requests by 40 percent. This server reduction contributed to Anzu’s cost savings and resulted in a 30 percent reduction in its carbon footprint. “We’re committed to cutting servers and not trees,” says Fenster.



Outcome | Using Amazon Bedrock to Enhance Game Advertisement

Within 3 months Anzu migrated to AWS and almost immediately achieved improved performance at a lower cost than with its previous cloud solution. Now, the company's internal teams can configure and scale networking infrastructure with ease. Additionally, Anzu's end customers benefit from faster bid requests and ad serving times, helping them reach gamers even more effectively.

Anzu plans to further reduce its carbon footprint by implementing [AWS Graviton processors](#), a family of processors designed to deliver the best price performance for cloud workloads running on Amazon EC2. The company has also finished a proof of concept for [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models from leading artificial intelligence companies, to use machine learning models for content moderation and improvements in contextual advertising. It plans to go live with these features within 3 months.

"Through our migration, we've found that AWS infrastructure is significantly better when it comes to meeting our networking, load balancing, and configuration needs," says Fenster. "We're seeing all kinds of improvements on AWS."

About Anzu

Anzu is an advanced intrinsic in-game ad solution for mobile, PC, and console gaming. The company works with developers and advertisers to create in-game ad experiences that bring value and don't interrupt gameplay.

AWS Services Used

Amazon CloudFront

Securely deliver content with low latency and high transfer speeds.

[Learn more »](#)

Elastic Load Balancing

Distribute network traffic to improve application scalability.

[Learn more »](#)



Amazon Elastic Kubernetes Service

Start, run, and scale Kubernetes without thinking about cluster management.

[Learn more »](#)

Amazon Elastic Computer Cloud (EC2) Spot Instances

Secure and resizable compute capacity for virtually any workload.

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